

Examiner reports

Q1.

- (a) This was well answered by the majority of candidates with many fully correct responses. The main error with the stretch was using the $\wedge$ -direction with a scale factor of 2 and with the translation the vector often contained +1.
- (b) This part was well answered by many candidates.
- (c)
 - (i) This was not very well answered by weaker candidates with many just rearranging the expression and then writing down the answer given without any convincing step being shown.
 - (ii) This was answered better than part c(i) with many correct factorisations seen. The main error was with the rejection of the solution $e^{2x} = -1$.
- (d) Some candidates obtained the correct answer and earned full marks. Most candidates made a reasonable attempt at the integrals with reasonable success and most showed enough to earn the B mark for subtraction of their areas. Incorrect limits of 6 and 0 often spoiled what would have been very good attempts.

Q2.

- (a)
 - (i) The vast majority of candidates found the partial fractions correctly, most substituting the two expected values of x . Some proceeded by setting up and solving simultaneous equations and some a hybrid of both methods, but few errors were made. A few lost all the marks here through having an algebraically incorrect opening line.
 - (ii) Here too, most candidates knew these were log integrals with relatively few showing no knowledge of this. The common error was to omit the '3 and a few candidates didn't have their log expressions in brackets and were penalised.
- (b) Various attempts at this question were seen, ranging from no attempt to a time consuming approach involving setting up simultaneous equations in P , Q and R following substitution of values of x , usually 0, 1 and 2. Most candidates who attempted this either set up just two equations, and thus found it was insoluble, or simply made errors in the algebra. The most successful and efficient method was algebraic long division. Relatively few candidates showed any knowledge of the technique of manipulating the numerator, as anticipated in the mark scheme. Some attempted to factorise either the numerator or denominator, or both, and made no further progress because these expressions did not factorise.

Q3.

- (a) Attempts at this question varied from no attempt, through superficial attempts to some fully correct solutions. Most could separate the variables correctly, although a few did attempt "integrals" of $f(x)dt$. Some candidates inverted the given differential equation and so could attempt direct integration, which was a quite acceptable method. Most integrated kdt correctly, but the integral of $(x+1)^{-\frac{1}{2}}$ defeated many.

Often there was an error in the power, before or after integration, and also in the coefficient. However, most candidates did include a constant and attempted to find it from the given initial conditions, for which they gained credit. However, some did some algebraic manipulation before adding a constant, and so could gain no credit, as this was then incorrect. Many candidates who had correctly integrated the expressions failed to gain the last mark, which was for giving the answer in the form $x = f(t)$, because they couldn't square their expression correctly, often omitting the "middle" term.

- (b) Candidates who had interpreted the context correctly could gain at least one mark here by substituting $t = 60$ into their result from part (a) and solving for x . However, many confused the given $t = 60$ with the given initial condition $x = 80$. Relatively few correct answers were seen, with many candidates obtaining results of over 100% or indeed negative results. These were not apparently queried, with little evidence of checking for errors.
- (c) (i) Very few candidates wrote down a correct differential equation. Some who had seemingly understood the context, wrote down the correct expression in A , but did not include the constant of proportionality. Many candidates mixed up the variables with x , and less so y , appearing, although most had a "differential equation" of some sort. Some candidates even included units in their differential equation, which was condoned if at the end of a correct expression.
- (ii) For many candidates this was the only part of this question, they attempted. Candidates who had understood the context and started with $A = 4.5$ could mostly solve the equation for t successfully. The relatively rare error was in the algebraic manipulation or in attempts to take logarithms, such as $\log(1 + 4e^{-0.09t}) = \log 1 - 0.09 \log 4t$ or similar. Some made an error in miscopying -0.09 .

Those candidates who commonly started with $A = 50$ or $A = 0.5$ encountered the need to take the logarithm of a negative number in their working or obtained a negative value for t , which most then chose to ignore. They gained no credit. Some did realise their error in interpreting the context, and reverted to the correct value for A .

©2025 Exam Papers Practice. All rights reserved.

Q4.

In part (a)(i) the majority of candidates realised the need to find the value of $f(x)$ when $x = 2$. However, it was also necessary, after showing that $f(2) = 0$, to write a statement that the zero value implied that $x - 2$ was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at Q . Those who attempted to find $\frac{dy}{dx}$ sometimes wrote it as $3x^2 + x$, but usually were aware of the need to substitute $x = 2$.

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in



finding the correct equation of the tangent, and most obtained at least a method mark here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some

wrote $\frac{x^4}{4} + \frac{x^2}{2} - 10$ and others gave $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$ as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate $6 - 20$ without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of -14 , the area of the shaded region was 14. A separate statement was needed and those who simply wrote $4 + 2 - 20 = -14 = 14$ did not score full marks. Those more able candidates who made a statement about the region being entirely below the x -axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

Q5.

This long calculus question with a final part covering transformations was a relatively high source of marks for many candidates. Most candidates were able to correctly differentiate

$2x^{\frac{3}{2}}$, but not all evaluated it correctly at $x = 4$. The majority of candidates showed a good understanding of the method involved in finding the equation of the normal in part (a)(ii), although some others did not use the correct gradient. Although the vast majority of

candidates knew the rules for integrating $8x^{\frac{1}{2}}$, there was a significant number of such candidates who did not simplify the coefficient correctly.

It was not uncommon to see either ' $\frac{8}{3}x^{\frac{3}{2}}$ ' not simplified or simplified incorrectly to ' $12x^{\frac{3}{2}}$ '.

Part (b)(ii) was generally answered better than expected, with a majority of candidates applying a correct method for finding the area of the shaded region. Although many candidates realised that the transformation was a 'translation', there were frequently sign errors in the vector, or components were interchanged. It was also disappointing to see a significant increase in the number of descriptions that involved more than one transformation which, as usual, was given no marks; the two transformations quoted were translation and a stretch.

Q6.

The full range of marks was seen in this question. Some candidates showed no understanding of how to proceed to solve the differential equation or how to use the given result to answer part (b). In contrast, some gave a concise and accurate derivation of the solution and understood how to use it in part (b). In part (a), algebraic error in the exponential term was common in attempted rearrangements, although some candidates did not separate x and t before attempting integration.

Similarly, in the integration of the exponential term, the sign or coefficient was often incorrect, although most candidates could integrate kt correctly. In part (a)(ii), most candidates had included a constant, and of these many changed its value whilst rearranging, by dividing by 2 or changing signs without saying so. Due to sign errors,

many could not take the logarithm of both sides correctly; although taking logarithms of a negative number was condoned for the method mark, $\ln(-kt^2)$ becoming $-\ln(kt^2)$ or similar was not. There was a lot of fudging of signs and coefficients in attempts to get to the given answer.

In part (b), candidates who went awry did not seem to realise they had been given the solution for x , and that the given 6000 matched the initial conditions given in part (a).

Many seemed to think that the 6000 was $\frac{dx}{dt}$, and that the given 10 was t , and they solved for x . Those who did write down the correct expression usually evaluated it correctly, with a few overlooking the request for the nearest hundred, this being a modelling situation. Relatively fewer candidates were successful in part (b)(ii). Some who had realised they were to solve for t , with x set to zero, did not deal with the 2 in the expression before exponentiation. A few candidates, who were otherwise correct, overlooked taking the square root in finding t .

Q7.

Most candidates gave the correct value for n in part (a). The most common wrong answer was ' $n = -\frac{1}{4}$ '.

The expansion of $\left(1 + \frac{3}{x^2}\right)^2$ in part (b) caused more difficulty than expected. A common wrong answer was ' $1 + 6x^{-2} + 9x^4$ ' although there were many other examples of incorrect answers offered.

In integrating $\left(1 + \frac{3}{x^2}\right)^2$ in part (c) most candidates realised that their expansion in part (b) should be used but surprisingly the integration of 1 with respect to x caused just as many problems as the integration of the terms in x raised to negative powers. Although many candidates showed that they knew how to deal with the given limits, a lack of brackets in their numerical expressions frequently led to the wrong answer. The final mark was frequently lost because no **exact** answer (fraction or mixed number) was given.

Q8.

Not all candidates appeared to have time to attempt this question and often solutions appeared rushed rather than thought out. However many of those who did attempt the question solved the differential equation in part (a) correctly. Most candidates separated the variables correctly using conventional notation, but then there were often errors in the integration. Common ones included the omission of a constant, incorrect integration of both x and $\cos 2t$ or use of degrees instead of radians when evaluating the constant. Some candidates just substituted the given conditions straight into the differential equation, producing nonsense.

In part (b)(i) some candidates substituted $\pi/4$ into dx/dt instead of their solution from part (a). Another error was to substitute $13\pi/4$ for t . There were, however, some correct

solutions showing understanding and most candidates who attempted this did show they intended to work with $\sin 26$, even if they were trying to use it in an incorrect expression or with their calculator in degrees. Most gave answers to the accuracy required by the question.

In part (b)(ii) those who had successfully solved the differential equation in part (a) usually earned the method mark here. However correct answers were rare, as many candidates arrived correctly at a negative value for $2t$, and just ignored the negative sign, rather than considering the next positive solution. Another common error lay in using the wrong calculator mode once more.

Q9.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that “ $x + 1$ is a factor”, after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote “integral = $-32 = 32$ ” gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate’s answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q10.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the

gradient of the normal in part (a)(iii) as -2 and so in part (a)(iv) obtained a positive value for the y -coordinate of Q . A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x -coordinate of the maximum point, M , was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their 'area under the curve' in part (b)(ii) of the question.

Q11.

Part (a)(i) was well answered.

Candidates who saw the connection with part (a)(i), usually made a very good attempt at part (a)(ii), often with complete success. Limits did cause a problem where candidates had not bracketed $2x^2 - 8x + 3$, and some candidates did not leave the answer in the required

form but left their answer as $\frac{1}{4} \ln 9$.

Part (b) was well answered by candidates who were able to write the integral in terms of u successfully. Many candidates lost a mark through omission of either du or c . A disturbing

number of candidates thought that $\frac{1}{3} \times \frac{1}{3} = \frac{1}{6}$. Some clearly able candidates lost the last mark by spoiling an otherwise acceptable answer.

Q12.

Some candidates gave a very concise and correct solution to this differential equation. Most candidates made some attempt to separate the variables and integrate, although it was not always clear that they had done this, with a stray x sometimes remaining on the left hand side. There were many errors in the integration, the common ones being $\ln y$, resulting from poor algebra in separation, and a coefficient error in the integral of $\cos 3x$, the 3 itself often being dropped and the integral given as $\sin x$.

Most candidates did include an arbitrary constant and tried to find it using the given conditions, although a few just substituted these into their solution without a $+ C$, or directly into the differential equation, and just produced nonsense.

Q13.

Part (a) was usually answered well and several correct equations were seen. However

errors such as $\frac{-15}{-3} = -5$ sometimes occurred, and weaker candidates confused coordinates and wrote equations such as $y - 15 = m(x + 2)$. Some attempted to differentiate to find the gradient instead of using the coordinates of the points.

Although the integration in part (b)(i) was largely successful, the evaluation of $F(1) - F(-2)$

defeated most. Arithmetic errors, usually caused by minus signs and fractions, abounded.

Some assumed that they had answered part (b)(ii) by substituting the limits after integration. Of those who realised the need to subtract the area of a triangle, many assumed the triangle base to be 4 rather than 3. Another error seen was the integration of the curve from -2 to $+2$ followed by subtracting their answer to part (b)(i). Some successfully found the area under the line by integration, but that was not the expected method.

Q14.

Although many candidates applied a correct method to find the expansion of $\left(1 + \frac{4}{x^2}\right)^3$, accuracy marks were sometimes lost because brackets were missing and the factor 4 was not squared when finding the value of q . Although a significant minority of candidates failed to use the expansion in part (a), despite the word 'Hence', most other candidates obtained the method mark in part (b), but it was surprising to find candidates having more difficulty integrating '1' than integrating the terms in x . Many candidates scored the method mark in part (b)(ii) but arithmetical errors were common in the subsequent evaluation.

Q15.

It was notable in this question that candidates tended to be successful in one part or the other but not both, although some good fully correct solutions were seen.

In part (a) many clear derivations of the given result were seen. These often involved some unnecessary steps, such as returning to double angles to eliminate $\cos^2 x$. Those who didn't complete this part successfully usually had an error in the double angle formulae, often omitting a 2. Many candidates wrote down the given answer even though this clearly did not follow from the previous line.

For part (b) most candidates knew they were expected to use the result from part (a) to do this integral; although some just ignored it, with $\sin^4 x$ often appearing in such solutions. Few correct alternative methods to the one expected were seen. The common errors in the expected method were in sign and coefficient errors in the integrals of $\sin x$ and $\sin 3x$.

Q16.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x -coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding

$\frac{1}{2} \times 1 \times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as -1 and the upper limit as 0. Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full

marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of -13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the

tangent was $\frac{-1}{17}$.

Q17.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$

instead of $\frac{dy}{dx} = 0$ dx^2 . In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating $(x + 1)$ were almost as

common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of the algebraic term $\frac{+4x^{-1}}{-1}$. In the final part, the evaluation of $F(4) - F(1)$ as 8.5 was the most common error.

Q18.

Part (a)(i) was well answered, with the main error being $\int -e^x dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4\ln 2 - 2 - \frac{1}{2} = 4\ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or -8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q19.

Most candidates just wrote down the required expression for $\cos^2 x$, although some worked it out from $\cos^2 x - \sin^2 x$ or $\cos(A + B)$. Omission of the 2 was a relatively rare error. Most candidates were able to derive the result in part (b)(i) although some were penalised for being too brief. Most candidates too were able to solve the resulting

quadratic equation, with some making it a little more difficult than intended by reading the 1 on the right hand side as 0. Most candidates found two results for angle x in the required range. Part (c) was less successful: although most candidates understood that they needed the integral in terms of $\cos^2 x$, there were many algebraic slips, and sign and coefficient errors in the integral of $\cos^2 x$ itself.

Q20.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at

the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^c$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t , rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.

Q21.

Part (a)(i) Most candidates realised the need to find the value of $f(x)$ when $x = 1$. However, it was also necessary, after showing that $f(1) = 0$, to write a statement that the zero value implied that $x - 1$ was a factor.

Part (a)(ii) Those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

Part (a)(iii) This section seemed unclear to some candidates. Many tried to find the discriminant but used the coefficients of the cubic equation. Many who used the quadratic thought that in order to have one real root the discriminant had to be zero, no doubt thinking the question was asking about equal roots. Some correctly stated that 1 was the only real root but many were obviously confused by the terms “factor” and “root” and stated that “ $x - 1$ was a root”.

Part (b)(i) Most candidates were well versed in integration and earned full marks here.

Part (b)(ii) The correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 1 and 2. Very few candidates realised the need

to find the area of a triangle as well and so failed to subtract the value of the integral from the area of the triangle in order to find the area of the shaded region.

Q22.

Most candidates were able to correctly simplify the given expressions although $x^{\frac{9}{4}}$ was a

common wrong answer in part (a)(iii) for the simplification of $\left(x^{\frac{3}{2}}\right)^2$. The majority of candidates displayed a good understanding of integration but a significant proportion of

the entry left their answer to part (b)(i) in the form $\frac{3}{2}x^{\frac{3}{2}}$. The final mark of three required the answer to be simplified further and the constant of integration present. The majority of candidates showed sufficient working in part (b)(ii) to show that they had used their answer to part (b)(i) and so gained the method mark, but a significant minority failed to

evaluate $2 \times 9^{\frac{3}{2}}$ correctly.

Q23.

Many candidates answered part (a) correctly and scored full marks but there were a significant number who gave a denominator of 10 in the second term of the answer after an otherwise correct approach. Some candidates tried to integrate x and differentiate e^{5x} which thereafter gave them no further credit.

In part (b)(i), most candidates obtained marks for finding $\frac{dz}{dx}$ but their attempts to obtain the

given answer were often flawed. Part (b)(ii) was not done particularly well; many split the denominator, and did not see that it required the use of $\ln$ at all.

©2025 Exam Papers Practice. All rights reserved.

Q24.

Part (a) was very well answered by the majority of candidates although errors of $\tan^2 x$ and $\ln(\sec x)$ were seen.

In part (b), the first step, the use of the quotient rule, was reasonably well done although the quotient rule was not always quoted correctly. Errors occurred where candidates had wrong signs for derivatives of $\sin x$ and $\cos x$ or where the functions in the numerator were reversed. Many candidates went on and used appropriate trigonometric identities to obtain the given answer.

Part (c) was not very well answered by most of the candidates. Many candidates obtained the correct expansion of $(\tan x + \cot x)^2$ but were then unable to handle the term $2 \tan x \cot x$. There were numerous flawed attempts using expressions such as $(\tan^2 x + 1) + (\cot^2 x - 1)$ equals the answer given.

In part (d), many candidates obtained the first method mark but went no further. Those who correctly integrated the expression often lost the final accuracy mark by working in degrees, with 57° being common.

Q25.

For part (a), most candidates knew that they were supposed to separate the variables and integrate, although interpretations of this varied widely. There were some impressive answers with clear integration leading to the final requested result. In contrast, some candidates were not sure what to do with dy and dx , these sometimes appearing as denominators. The weakest candidates simply substituted the given values of x and y into the differential equation and produced meaningless nonsense. Most candidates who had

separated correctly at least got as far as $k\sqrt{1+2y}$ for their y integral, although an index of $\frac{3}{2}$ or $-\frac{1}{2}$ was a fairly common error.

It was pleasing to see the integration of $\frac{1}{x^2}$ done correctly by many candidates, with the common error being in the index. It was also pleasing that most candidates included an arbitrary constant in their solution and tried to find it using the given condition.

In part (b), most candidates who had $k\sqrt{1+2y}$ in their solution realised that they were to square it to move towards the given result, but many attempts were poor, with

$(f(x)+c)^2 = (f(x))^2 + c$ being frequently seen, despite there being a term in $\frac{1}{x}$ in the given solution. Some candidates who had correctly introduced a constant c in part (a) proceeded to manipulate their solution without including c in the manipulation, and thus failed to score the mark for finding it.

Q26.

(a) Not everyone recognised that the height of the rectangle was the value of y when $x = -1$ or $x = 4$. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.

(b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.

(ii) Almost everyone recognised that they should firstly evaluate the integral from -1 to 2 , but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it

correctly.

- (c) (i) Differentiation was done well on the whole.
- (ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.
- (iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.

Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.

- (d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$ as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as -2 , $\sqrt{2}$, $1 + \sqrt{2}$. Using the formula or completing the square sometimes led to $1 \pm \sqrt{1}$, which many candidates failed to simplify.

Q27.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P . A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle

then using the cosine rule to find angle OPA then using the formula $\frac{1}{2} ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more

directly by using $\frac{1}{2} OA \times |y_P|$.

Q28.

Part (a) was usually correct. Some candidates just wrote the result down whilst others

derived it from $\cos^2 x - \sin^2 x$. The common error was to have $2 \cos^2 x + 1$, or the 2 missing. There were many fully correct responses to part (b) and nearly all candidates made some attempt to make use of the double angle in their integration. Accuracy sometimes slipped with the $1/2$ attaching to the $\cos 2x$ term only. Virtually all candidates gave evidence of correct use of limits, but those who didn't obtain the requested form of

$\frac{\pi}{a}$, showed no attempt to find their error.

Q29.

Part (a)(i) Most candidates differentiated correctly. However a few made a slip or misread one of the terms.

Part (a)(ii) Although most managed to substitute 2 into their derivative some made numerical errors and some used y or the second derivative. Most who realised that they should equate their derivative to zero (or at least showed their intention though never inserting the $= 0$) then tried to factorise or use the formula (though it was clear that some did not recognise the quadratic equation as such). It was disappointing that 14 the bracket $(3x - 14)$ often produced the solution $x = 14$ instead of $14/3$, and the solution of $x = 2$ did not always appear.

Part (b)(i) The integration was also completed correctly by most candidates, although the

$28x$ was occasionally wrong and some 'hybrid' processes led to terms such as $\frac{20x^3}{3}$

In part (b)(ii) almost everyone attempted to substitute 3 into their integral but their problems with the ensuing fractions often took pages to resolve, and although most ended 'magically' with the required answer there were many errors en route. A few substituted into the original expression for y instead of the integrated expression.

Part (b)(iii) This part was quite well done although again there were errors in evaluating

both $\frac{1}{2} \times 21 \times 3$ and $56 \frac{1}{4} - 31 \frac{1}{2}$. Some candidates confused length with area and merely used Pythagoras's Theorem to find the length of the hypotenuse of the triangle. Those who chose to integrate the equation of the straight line were sometimes successful but many made arithmetic errors.

Q30.

This question also proved to be a good source of marks for many candidates. The examiners expected to see some intermediate evaluation before the printed answer was quoted in part (a)(i) after the substitution of 4 for x in the given equation. Most candidates gave the correct value for the power k although the wrong values $1/2$ and $1/2$ were both seen in part (a)(ii). Candidates were generally able to differentiate the given expression to find the second derivative in part(a)(iii) but some believed that the sign of the x value rather than the sign of the second derivative determined the nature of the stationary point in (iv). In part (b) a significant number of candidates gave the equation of the tangent instead of the normal or used the gradient of the normal as -12 . Finding the integral in part (c)(i) was well answered although, somewhat surprisingly, the integration of -7 caused as many problems as the integration of the other two terms. The final part of the

question proved to be beyond many average candidates who forgot to insert the constant of integration in answering the previous part. The hint 'Hence', which should have suggested the use of the answer to (c)(ii) was not always picked up. It was not uncommon to see the equation of the curve given as a linear equation, normally the tangent at P .

Q31.

There were many good answers to this question, with full or nearly full marks.

Part (a) Partial fractions is an area that is generally known well, although the presence of the 3 led some candidates to make an error, some ignoring it altogether, and others not multiplying it by the common denominator. Some candidates simplified the given expression by dividing it out, which was perfectly valid. Those candidates who substituted $x = 1$ and $x = -$ usually found the coefficients correctly; many candidates who chose to set up simultaneous equations by equating coefficients made an error.

Part (b) Virtually all candidates knew they were to use the partial fractions to do the integral and that it involved $\ln(3x - 1)$ and $\ln(x - 1)$, and very few nonsense "integrations" were seen. The common errors were in the coefficients with the $1/3$ from $3x$ being missed, or the coefficients given as fractions $1/6$ and $1/4$.

Q32.

Many full mark answers were seen to this question with candidates demonstrating confidence in solving the equation. Some lost the last mark, or two, through incorrect manipulation to the requested $y = f(x)$ form. Finding the value of the constant after

inverting to $y = -\frac{1}{3x^2}$ was the common error. Poorer attempts showed no knowledge of the separation of variables technique, and candidates attempted to use integration by parts. Others made errors in the algebra, or in the integration or did not include a

constant. Integrating $\frac{1}{y^2}$ to an expression involving a $\ln$ function was fairly common.

©2025 Exam Papers Practice. All rights reserved.

Q33.

- (a) (i) Some were confused by the word "factor" and gave answers as 1 and -2 .
- (ii) Although there were many correct answers given for the factorisation, some candidates appeared to be guessing the factors of $f(x)$, making no use of the information given in part (a)(ii). Despite getting a non-zero remainder for part (a)(i), several candidates gave $(x + 1)$ as a factor.
- (b) Most candidates found the coordinates of A , and those with the correct factor $(x - 4)$ of $f(x)$ were usually able to find the correct coordinates of B .
- (c) It was pleasing to see most candidates able to integrate correctly, although quite a few failed to integrate 8 to give $8x$. Some candidates did not evaluate the limits in the correct order, but there were many correct values seen for the area of the shaded region.

Q34.

Parts (a) and (b) were answered very well and although the powers in answers to part (c) were nearly always correct, the coefficients were sometimes the reciprocal of the powers in part (b). Part (d) caused significantly more problems to the candidates. Although the method of solving a definite integral was well understood, many candidates resorted to using decimal approximations and, in general, candidates displayed a weakness in converting fractional powers of numbers back into surds.

Q35.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted $x = 3$ into the given cubic equation and found $p(x)$ to be zero. Some, however, lost a mark as they did not indicate that $p(3) = 0$ implied that $x - 3$ was a factor.

Part (a)(ii) The coordinates $(3, 0)$ were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 and $7/3$. Many found the second derivative in order to ascertain which point was the minimum point M . This was unnecessary as the coordinates of A $(1, 0)$ were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of $x = 1$.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was $11/12$ and the negative value of the integral arose because the region was below the x -axis. It was not acceptable to simply

write Area = $-\frac{11}{12} = \frac{11}{12}$.